

Ameya Panchal

State College, PA • ameyapanchal011@gmail.com • (781) 870-7474 • linkedin.com/in/ameya-bit • github.com/Ameya-bit

Education

The Pennsylvania State University

Bachelor of Science in Computer Science; Minor in Physics

University Park, PA

Expected May 2028

- GPA: 3.76/4.00 | Schreyer Honors College Scholar
- Relevant Coursework: Data Structures & Algorithms, Systems Programming, Physics (Mechanics/E&M), Discrete Math

Experience

The Pennsylvania State University

Undergraduate Research Assistant (AI & Computational Modeling)

State College, PA

Oct 2025 – Present

- Spearhead independent research on Monte Carlo simulations, integrating AI models to enhance probabilistic forecasting accuracy by 97%.
- Engineer complex computational models in Python to optimize statistical simulation methodologies, reducing processing time for large datasets.
- Analyze experimental data using Scikit-learn to identify patterns and support breakthroughs in statistical modeling.

Penn State Office of the Physical Plant

Part-Time Application Specialist

State College, PA

Dec 2025 – Present

- Researched and documented 2 API protocols (REST/SOAP) for the Docfinity system, reducing onboarding time for new developers by 30%.
- Analyzed 5+ metadata extraction strategies from the AI pipeline documentation and identified 3 key edge cases in the existing codebase.
- Collaborated with 4 cross-functional team members across 3 weekly meetings to prepare technical documentation for product launch.

Outlier

Generative AI Trainer

Remote

Aug 2024 – Present

- Review and edit Python and JavaScript code for front and back-end AI projects to ensure correct library usage and code structure.
- Boost model accuracy and reliability by providing rigorous technical feedback on AI-generated code snippets.

Leadership & Activities

Penn State University

Physics Learning Assistant

State College, PA

Aug 2025 – Dec 2025

- Collaborated with professors to support students in understanding physics concepts, providing guidance on homework, labs, and exams.
- Facilitated group discussions and fostered an inclusive learning environment for undergraduate physics students.

College of IST Living Learning Community

Vice President

State College, PA

Aug 2025 – Present

- Spearheaded executive board initiatives to drive member engagement, resulting in a 20% increase in active participation.
- Oversaw event logistics and leveraged scheduling software to ensure seamless operations for multiple participant groups.

Skills & Interests

Technical: Python, Java, JavaScript, React.js, Flask, TensorFlow, Scikit-learn, SQL, Git/GitHub

Laboratory/Research: Monte Carlo Simulations, Data Analysis, Biomedical Signal Analysis, Research Methodology

Languages: English (Native)